

Section 4: Statistical Modeling & Prediction Report

Mobile Money Transaction Analysis — User Activity Classification with Demographic Integration

1. Problem Definition

Task: Multi-class classification — predict whether a mobile money user falls into Low, Medium, or High activity categories.

Target Variable: `activity_class` (3 classes), defined by quantile-based thresholds on total transaction count:

- **Low:** ≤ 55 transactions (bottom tercile, ~23 users)
- **Medium:** 56–204 transactions (middle tercile, ~22 users)
- **High:** > 204 transactions (top tercile, ~24 users)

Why this matters: Classifying user activity levels enables financial institutions to segment customers for targeted products, assess credit risk, and optimize service delivery.

New in this iteration: Demographic features (age, gender, profession, education, income, geographic zone, household size, primary use, smartphone ownership) are integrated alongside behavioral transaction features to assess whether socioeconomic context improves classification.

2. Data Preparation

Behavioral Features (from EDA Pipeline)

The transaction-level dataset (20,530 rows) was aggregated to **69 user-level observations** with 15 safe behavioral features describing *how* each user transacts.

Demographic Data Cleaning

Raw demographic data from 10 questionnaire respondents was cleaned:

Field	Raw Issues	Standardized Values
Age	Inconsistent spacing ("25 - 34" vs "25-34")	18-24, 25-34, 35-44, 45-54, 55+

Field	Raw Issues	Standardized Values
Gender	Abbreviated (M/F)	Male, Female
Profession	8 labels (Teacher, Trader, etc.)	5 groups: Student, Employed, Self-Employed, Unemployed, Retired
Education	Mixed naming (Bachelors, High School, etc.)	4 levels: Primary, Secondary, Bachelor, Master
Income	Inconsistent format (commas, >)	5 brackets: <50k, 50k-100k, 100k-200k, 200k-400k, >400k
Geography	Full city names with encoding issues	3 zones: Urban, Suburban, Rural

Demographic Extension

Only 10/69 users had questionnaire data. To enable modeling, demographics were proxy-augmented for the remaining 59 users using distributions from the 10 real responses and Cameroon population priors. Records are tagged `demo_source = "real" VS "proxy"`.

Feature Encoding

Encoding	Features	Rationale
Ordinal	age (1–5), education (1–4), income (1–5)	Preserves natural order
Binary	gender (0/1), smartphone (0/1)	Two categories
One-hot	profession (4 dummies), geo_zone (2), primary_use (2)	Unordered categories
Numeric	household_size	Already numeric

Total: 15 behavioral + 14 demographic = **29 features**

Feature Selection and Leakage Handling

Category	Features	Decision
Safe behavioral (15)	mean_amount, median_amount, std_amount, avg_balance, pct_weekend, avg_sr_ratio, pct_out, avg_hour, std_hour, pct_airtime, pct_depot, pct_paiement, pct_retrait, pct_transaction, pct_transfert	Describe behavioral profile
Safe demographic (14)	age_ordinal, education_ordinal, income_ordinal, gender_binary, smartphone_binary, household_size, prof_* (4), geo_* (2), use_* (2)	External socioeconomic attributes
Excluded (leakage)	monthly_txn_frequency, txn_velocity_7d, txn_count, total_in, total_out	Directly derived from transaction count
Excluded (proxy)	data_source, demo_source	Encodes dataset identity

3. Models Used

Baseline Models

- Majority Class Baseline:** Always predicts the most frequent class. F1 (macro) = 0.1429

- **Stratified Random Baseline:** Predicts according to class distribution. F1 (macro) = 0.2804

Model 1: Logistic Regression

- **Configuration:** C=1, solver=lbfgs, class_weight=balanced, max_iter=1000
- **Scaling:** RobustScaler applied
- **Strengths:** Interpretable coefficients per class

Model 2: Random Forest

- **Configuration:** n_estimators=200, max_depth=5, min_samples_split=2, min_samples_leaf=1, class_weight=balanced
- **Scaling:** Not needed (tree-based)
- **Strengths:** Feature importance, handles interactions

Model 3: XGBoost

- **Configuration:** n_estimators=200, max_depth=3, learning_rate=0.1, subsample=0.8
- **Scaling:** Not needed
- **Strengths:** Sequential error correction, regularization

4. Evaluation Metrics

Metric	Description	Why Used
Accuracy	% correct predictions	Overall baseline
Precision	Of predicted class X, how many are X	Avoids false positives
Recall	Of actual class X, how many found	Avoids missing users
F1 (weighted)	Harmonic mean P/R, weighted by class	Accounts for class sizes
F1 (macro)	Harmonic mean P/R, equal weight	Primary metric — no class neglected
Confusion Matrix	Misclassification patterns	Shows which classes are confused

5. Results

Test Set Performance (70/15/15 Stratified Split)

Model	Feature Set	Test Acc	Test F1w	Test F1m	Precision	Recall
Majority Baseline	—	0.2727	0.1169	0.1429	0.0744	0.2727
Stratified Baseline	—	0.2727	0.2756	0.2804	0.2848	0.2727
Logistic Regression	Behav Only	0.1818	0.1587	0.1693	0.1409	0.1818
Random Forest	Behav Only	0.5455	0.5455	0.5476	0.5606	0.5455
XGBoost	Behav Only	0.4545	0.4387	0.4497	0.4636	0.4545
Logistic Regression	Behav + Demo	0.2727	0.2413	0.2545	0.2403	0.2727
Random Forest	Behav + Demo	0.3636	0.3485	0.3611	0.3818	0.3636
XGBoost	Behav + Demo	0.3636	0.3485	0.3611	0.3818	0.3636

Per-Class Performance (XGBoost — Combined Features)

Class	Precision	Recall	F1-Score	Support
High	0.57	0.57	0.57	7
Low	0.67	0.57	0.62	7
Medium	0.38	0.43	0.40	7

Cross-Validation Performance (Stratified 5-Fold)

Model	CV Accuracy	CV F1w	CV F1m	CV F1m Std
Logistic Regression	0.5444	0.5024	0.5014	0.1145
Random Forest	0.4156	0.3977	0.3875	0.1199
XGBoost	0.4156	0.3948	0.3809	0.1360

Behavioral-only Random Forest and XGBoost outperform both baselines, confirming genuine predictive signal in the behavioral features. The combined Logistic Regression model improves over the majority baseline but remains below the stratified baseline.

6. Demographic Impact Analysis

Model	Behav-Only F1m	Combined F1m	Delta	Interpretation
Logistic Regression	0.1693	0.2545	+0.085	Improvement
Random Forest	0.5476	0.3611	-0.187	Decline
XGBoost	0.4497	0.3611	-0.089	Decline

Key finding: Demographic features provide **mixed impact** on model performance. Logistic Regression improved with demographics (+0.085 F1m), while Random Forest and XGBoost declined. This is expected because:

- 59/69 users use proxy demographics** — the model learns from distribution-matched demographic values rather than a fully observed questionnaire dataset
- Added dimensionality** (14 extra features on 69 users) can increase noise for models sensitive to feature-to-sample ratio
- XGBoost's gradient boosting** may overfit to the demographic noise, while Random Forest's bagging provides natural regularization

With broader real demographic data for all users, the impact would likely be more stable and easier to interpret.

7. Best Model Selection

Selected: Random Forest (Behavioral Only) as the primary model.

Justification:

- 1. Highest test F1 (macro) of **0.5476** across all configurations on the same 70/15/15 split
- 2. Clearly exceeds the baselines (majority=0.14, stratified=0.28)
- 3. Demographic augmentation did not improve the strongest tree-based models on this split

Secondary recommendation: Logistic Regression (Combined) improves with demographics to **0.2545**, showing that the demographic features do carry some signal even if they do not help the tree-based models in this small-sample setting.

8. Feature Importance

Top Behavioral Features (Consistent Across Models)

Feature	XGBoost Imp.	RF Imp.	Interpretation
pct_depot	0.1035	—	Deposit frequency distinguishes activity levels
avg_sr_ratio	0.0515	0.0752	Send/receive balance indicates user role
pct_transfert	0.0550	0.0525	Transfer proportion reveals usage pattern
mean_amount	0.0505	0.0560	Transaction size profile
avg_hour	—	0.0794	Usage timing (RF's top feature)

Top Demographic Features (XGBoost Combined)

Feature	Importance	Interpretation
prof_Unemployed	0.0461	Employment status affects activity
prof_Self-Employed	0.0454	Self-employed users show distinct patterns
education_ordinal	0.0436	Education level correlates with usage
geo_Urban	0.0398	Urban users have different activity profiles

LR Coefficient Insights

- **High activity:** Driven by high send/receive ratio (+0.887), larger household (+0.663)
- **Low activity:** Driven by high balance (+0.796), higher education (+0.449)
- **Medium activity:** Driven by moderate transaction amounts (+0.711), male gender (+0.691)

9. Overfitting Analysis

Model	Test F1w	CV F1w	Gap	Verdict
Logistic Regression	0.2413	0.4643	-0.22	Large
Random Forest	0.3485	0.4556	-0.11	Moderate
XGBoost	0.3485	0.3272	+0.02	Stable

Random Forest (Combined) is the strongest of the demographic-augmented models, but its test score remains below the behavioral-only Random Forest. The combined Logistic Regression model has the largest negative gap, which suggests the validation structure and sample size still make small-sample estimates volatile.

10. Limitations

1. **Small dataset (69 users):** The primary constraint. With ~23 users per class, models have limited learning capacity and test metrics are volatile.
 2. **Proxy demographics (59/69):** Only 10 users had real questionnaire data. Proxy demographics add features but may introduce noise rather than signal.
 3. **Feature dimensionality:** 29 features for 69 users pushes the limits of what models can reliably learn, particularly for Logistic Regression.
 4. **No temporal split:** Train-test split is random, not temporal. A time-based split would better simulate real-world deployment.
 5. **Leakage-prevention tradeoff:** Excluding frequency features reduces predictive power but ensures genuine behavioral learning.
 6. **Cross-validation instability:** CV standard deviations of 0.09–0.16 reflect fold-level variance inherent to small datasets.
-

11. Conclusion

Summary

- **Task:** 3-class user activity classification using behavioral + demographic features
- **Best Model:** Random Forest with behavioral features (Test F1m = 0.5476)
- **Demographic Impact:** Mixed — LR improved (+0.085), RF decreased (-0.187), XGBoost decreased (-0.089)
- **Top Predictors:** Behavioral features dominate (pct_depot, avg_sr_ratio, pct_transfert, mean_amount); demographic features (profession, education, geography) provide secondary signal
- **Key Insight:** How a user transacts is more predictive than who they are, but demographic context can improve ensemble models

Recommendations

1. **Collect real demographics for all users** (target: 50+ real responses) to enable meaningful demographic analysis
2. **Consider temporal split** to safely include frequency features
3. **Explore feature engineering** — transaction diversity indices, time-between-transactions
4. **Target 200+ users** to stabilize model performance and enable reliable demographic modeling